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# BE THE GREEN CHAMPION

**India's success in building up its renewable energy base and the successful dissemination of this knowledge to other countries will help it consolidate its position as a global leader in the sector**

**In 2023, India added 17,960 MW of electricity capacity; 72.3 per cent of it was renewable energy.** Today, in India (as in most other geographies), solar electricity is the cheapest form of electricity available—though only when the sun is shining. This highlights the essential conundrum about solar and wind energy, especially in developing countries where the energy demand is still increasing both during the daytime and night hours.

Consequently, as we plan ahead, the key question is whether we add renewable energy capacity alone (thus depriving people of electricity at night) or add both solar capacity and fossil fuel-based capacity for the times when the sun is not shining and the wind is not blowing (thus increasing the cost of electricity for all)? In the developed countries, by contrast, no additional electricity capacity needs to be added; and solar and wind energy are backed by existing power capacity when solar and wind energy are not available.

In the progressive policy environment, we have, very recently—in the past six months or so—found prices, through tenders, for round-the-clock renewable energy (RTC-RE) at 85 per cent and 90 per cent availability (similar to that of fossil fuel-based power stations), based on solar + wind capacity (and sometimes with battery capacity to enable the achievement of higher quality), to be Rs 4 to 5 per kWh (kilowatt hour), whereas new coal-based power stations have been awarded tariffs in excess of Rs 5 per kWh. This is a game-changer, in as much as RTC-RE is now cost-effective in providing baseload power. And renewable electricity could be supplemented (by much more than only the balance 10-15 per cent availability requirement) by nuclear power, hydro power and other zero carbon-emitting electricity sources.

This highlights India's success in building up its renewable energy base. No other country—especially a developing one—has been able to meet the challenge of growing demand and RTC-RE supply at competitive prices. This also provides the basis for India to become a global champion for renewable electric power. For example, the policies that we have implemented and the lessons that we have learned have, time and again, shown us the way ahead to tweak and make the policies that result in even lower costs. This promotes and

## THE GIANT STEPS

- Solar electricity is the cheapest form of electricity available in India today, though only when the sun is shining
- India needs to figure out whether it adds renewable energy capacity alone or also fossil fuel–based capacity for when the sun is not shining and the wind is not blowing
- India is the only developing country that has been able to meet the challenge of growing demand and round-the-clock renewable electricity supply at competitive prices